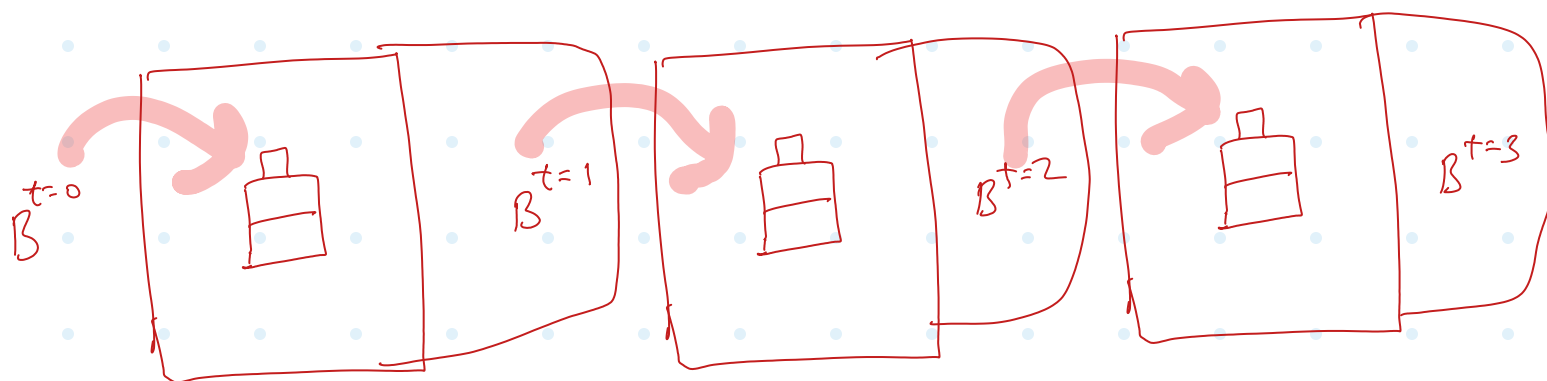




$$f(n^{t=1})$$

s.t.:

$$g(n^{t=1}) \leq 0$$

$$B^{t=1} - \bar{B} \leq 0$$

$$\underline{B} - B^{t=1} \leq 0$$

$$h(n^{t=1}) = 0$$

$$B^{t=1} - B^{t=0} + \Delta t P_B^{t=1} = 0$$

$$f(n^{t=2})$$

s.t.:

$$g(n^{t=2}) \leq 0$$

$$B^{t=2} - \bar{B} \leq 0$$

$$\underline{B} - B^{t=2} \leq 0$$

$$h(n^{t=2}) = 0$$

$$B^{t=2} - B^{t=1} + \Delta t P_B^{t=2} = 0$$

$$f(n^{t=3})$$

s.t.:

$$g(n^{t=3}) \leq 0$$

$$B^{t=3} - \bar{B} \leq 0$$

$$\underline{B} - B^{t=3} \leq 0$$

$$h(n^{t=3}) = 0$$

$$B^{t=3} - B^{t=2} + \Delta t P_B^{t=3} = 0$$

MPOF Problem:

$$\sum_{t_0=1}^3 f(n^{t=t_0})$$

s.t.

$$g(n^{t=t_0}) \leq 0$$

$$h(n^{t=t_0}) = 0$$

$$B^{t=t_0} - \bar{B} \leq 0$$

$$\underline{B} - B^{t=t_0} \leq 0$$

$$B^{t=t_0} - B^{t=t_0-1} + \Delta t P_B^{t=t_0} = 0$$

$$t_0 = 1:3$$

$$\begin{aligned}
 L_{\text{MPOPF}} = & \sum_{t=1}^3 \left\{ f(n^{t=t_0}) + \lambda^{t=t_0} \cdot g(n^{t=t_0}) \right. \\
 & \left. + \mu^{t=t_0} \cdot h(n^{t=t_0}) \right\} \\
 & + \sum_{t=1}^3 \left\{ \lambda_{\underline{B}}^{t=t_0} (\underline{B} - B^{t=t_0}) \right. \\
 & \left. + \lambda_{\overline{B}}^{t=t_0} (B^{t=t_0} - \overline{B}) \right\} \\
 & + \sum_{t=1}^3 \mu_{\text{soc}}^{t=t_0} (B^{t=t_0} - B^{t=t_0-1} + \Delta t P_B^{t=t_0})
 \end{aligned}$$

IO condition: For some optimal point $y^* = (n^*, B^{t=1}, B^{t=2}, B^{t=3})$

① $\nabla_{y^*} L_{\text{MPOPF}} = 0 \Rightarrow \nabla_{B^{t=1}} L_{\text{MPOPF}} = 0$

$\nabla_{B^{t=2}} L_{\text{MPOPF}} = 0$

$\nabla_{B^{t=3}} L_{\text{MPOPF}} = 0$

$$\begin{aligned}
 \nabla_{B^{t=1}} L_{\text{MPOPF}} &= \left[-\lambda_{\underline{B}}^{t=1} + \lambda_{\overline{B}}^{t=1} + \mu_{\text{soc}}^{t=1} - \mu_{\text{soc}}^{t=2} = 0 \right. \\
 \nabla_{B^{t=2}} L_{\text{MPOPF}} &= \left[-\lambda_{\underline{B}}^{t=2} + \lambda_{\overline{B}}^{t=2} + \mu_{\text{soc}}^{t=2} - \mu_{\text{soc}}^{t=3} = 0 \right. \\
 \nabla_{B^{t=3}} L_{\text{MPOPF}} &= \left[-\lambda_{\underline{B}}^{t=3} + \lambda_{\overline{B}}^{t=3} + \mu_{\text{soc}}^{t=3} = 0 \right]
 \end{aligned}$$

② Convert dual inequality of $\lambda_{\underline{B}}^{t=t_0} \cdot (\underline{B} - B^{t=t_0}) \leq 0 \Rightarrow \lambda_{\underline{B}}^{t=t_0} \geq 0$

$$\lambda_{\bar{B}}^{t=t_0} \cdot (B^{t=t_0} - \bar{B}) \leq 0 \Rightarrow \boxed{\lambda_{\bar{B}}^{t=t_0} \geq 0}$$

③

$$\mu_{\text{soc}}^{t=t_0} (B^{t=t_0} - B^{t=t_0-1} + \Delta t P_B^{t=t_0}) = 0$$

Equality constraint.

$$\Rightarrow \boxed{\mu_{\text{soc}}^{t=t_0} \in \mathbb{R}}$$

④

Complementary slackness

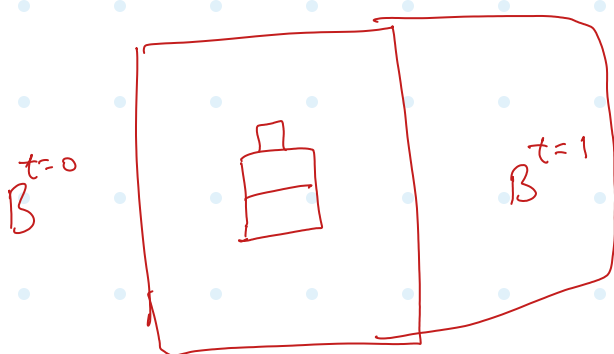
$$\lambda_{\underline{B}}^{t=t_0} = 0 \quad \Rightarrow \quad \underline{B} - B^{t=t_0} < 0$$

$$\lambda_{\bar{B}}^{t=t_0} = 0 \quad \Rightarrow \quad B^{t=t_0} - \bar{B} < 0$$

Again:

$$\begin{aligned} L_{\text{MPOPF}} = & \sum_{t=1}^3 \left\{ f(u^{t=t_0}) + \lambda^{t=t_0} \cdot g(u^{t=t_0}) \right. \\ & \left. + \mu^{t=t_0} \cdot h(u^{t=t_0}) \right\} \\ & + \sum_{t_0=1}^3 \left\{ \lambda_{\underline{B}}^{t=t_0} (\underline{B} - B^{t=t_0}) \right. \\ & \left. + \lambda_{\bar{B}}^{t=t_0} (B^{t=t_0} - \bar{B}) \right\} \\ & + \sum_{t_0=1}^3 \mu_{\text{soc}}^{t=t_0} (B^{t=t_0} - B^{t=t_0-1} + \Delta t P_B^{t=t_0}) \end{aligned}$$

$$L_{\text{MPOPF}} = \sum_{t=1}^{t=t_0} L_{\text{ACOPF}}$$



$$f(n^{t=1})$$

s.t.:

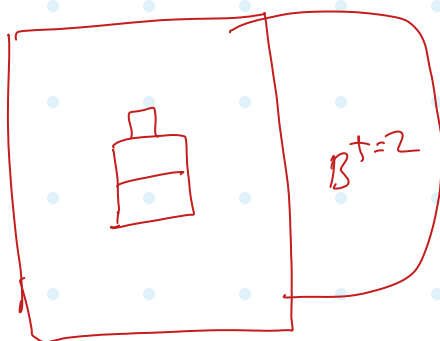
$$g(n^{t=1}) \leq 0$$

$$B^{t=1} - \bar{B} \leq 0$$

$$\underline{B} - B^{t=1} \leq 0$$

$$h(n^{t=1}) = 0$$

$$B^{t=1} - B^{t=0} + \Delta t P_B^{t=1} = 0$$



$$f(n^{t=2})$$

s.t.:

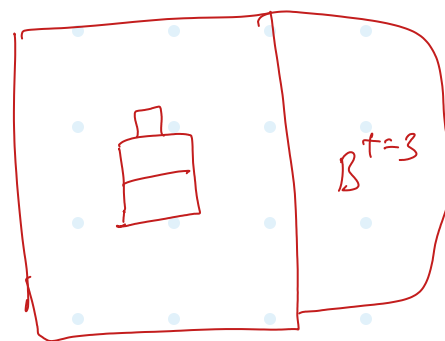
$$g(n^{t=2}) \leq 0$$

$$B^{t=2} - \bar{B} \leq 0$$

$$\underline{B} - B^{t=2} \leq 0$$

$$h(n^{t=2}) = 0$$

$$B^{t=2} - B^{t=1} + \Delta t P_B^{t=2} = 0$$



$$f(n^{t=3})$$

s.t.:

$$g(n^{t=3}) \leq 0$$

$$B^{t=3} - \bar{B} \leq 0$$

$$\underline{B} - B^{t=3} \leq 0$$

$$h(n^{t=3}) = 0$$

$$B^{t=3} - B^{t=2} + \Delta t P_B^{t=3} = 0$$

$$\begin{aligned} L_{\text{ACOPF}} &= f(n^{t=1}) \\ &+ \lambda^{t=1} g(n^{t=1}) \\ &+ \mu^{t=1} h(n^{t=1}) \end{aligned}$$

$$+ \lambda \frac{t=1}{B} (B^{t=1} - \bar{B})$$

$$+ \lambda \frac{t=1}{B} (\underline{B} - B^{t=1})$$

$$+ \mu_{\text{soc}}^{t=1} (B^{t=1} - B^{t=0} + \Delta t P_B^{t=1})$$

$$\begin{aligned} L_{\text{ACOPF}} &= f(n^{t=2}) \\ &+ \lambda^{t=2} g(n^{t=2}) \\ &+ \mu^{t=2} h(n^{t=2}) \end{aligned}$$

$$+ \lambda \frac{t=2}{B} (B^{t=2} - \bar{B})$$

$$+ \lambda \frac{t=2}{B} (\underline{B} - B^{t=2})$$

$$+ \mu_{\text{soc}}^{t=2} (B^{t=2} - B^{t=1} + \Delta t P_B^{t=2})$$

$$\begin{aligned} L_{\text{ACOPF}} &= f(n^{t=3}) \\ &+ \lambda^{t=3} g(n^{t=3}) \\ &+ \mu^{t=3} h(n^{t=3}) \end{aligned}$$

$$+ \lambda \frac{t=3}{B} (B^{t=3} - \bar{B})$$

$$+ \lambda \frac{t=3}{B} (\underline{B} - B^{t=3})$$

$$+ \mu_{\text{soc}}^{t=3} (B^{t=3} - B^{t=2} + \Delta t P_B^{t=3})$$

$$\nabla_{\underline{B}^{t=1}} L^{t=1}_{\text{ACORF}} = 0 = \underbrace{\lambda_{\underline{B}}^{t=1} - \lambda_{\underline{B}}^{t=1} + \mu_{\text{soc}}^{t=1}}_{\text{NOT MP-optimal}} = 0$$


inbuilt
in
 $t=1$
 L_{ACORF}

$$\boxed{-\lambda_{\underline{B}}^{t=1} + \lambda_{\underline{B}}^{t=1} + \mu_{\text{soc}}^{t=1} - \mu_{\text{soc}}^{t=2} = 0} \quad \text{But this is MP-optimal} \quad \text{✱}$$

So let's nudge $\text{ACORF}^{t=1}$ problem so that we end up forcing ✱ instead of the in-built 

We wanna add

$$- \mu_{\text{soc}}^{t=2} L(\underline{B}^{t=1})$$

↓
a linear function
in $\underline{B}^{t=1}$

$$L_{\text{AcoPr}}^{t=1} = f(n^{t=1}) \\ + \lambda^{t=1} g(n^{t=1}) \\ + \mu^{t=1} h(n^{t=1})$$

$$+ \lambda_{\bar{B}}^{t=1} (\bar{B}^{t=1} - \bar{B})$$

$$+ \lambda_{\underline{B}}^{t=1} (\underline{B} - \underline{B}^{t=1})$$

$$+ \mu_{\text{soc}}^{t=1} (\bar{B}^{t=1} - \bar{B}^{t=0} + \Delta t p_B^{t=1})$$

$$+ \mu_{\text{soc}}^{t=2} (\bar{B}^{t=2} - \bar{B}^{t=1} + \Delta t p_B^{t=2})$$

$\Rightarrow$

$$\min \quad f(n^{t=1}) - \mu_{\text{soc}}^{t=2} \bar{B}^{t=1}$$

s.t.

$$g(n^{t=1}) \leq 0$$

$$h(n^{t=1}) = 0$$

$$\bar{B}^{t=1} - \bar{B}^{t=0} + \Delta t p_B^{t=1} = 0$$

$$\bar{B}^{t=1} \in [\underline{B}, \bar{B}]$$

$$p_B^{t=1} \in [-p_{Br}, p_{Br}]$$

